

Assignment #3

Deliverables:

Your assignment submission ZIP file ("HW3_YourLastName")) must have the following items in it:

1. Your **ZIP** file must have two directories: **src** and **doc**
2. Your **src** directory must have 2 subdirectories: **Part_I** and **Part_II**
3. Your **Readme** files must have the detailed steps to install, deploy, and run every IPYNB script and required packages
4. Use **Panopto** to create 5-10 minutes **video** for a live-demo of your application
5. All source code must be stored under **src** directory
6. Documentation and video must be stored under **doc** directory

PART I (100 points):

Create a Python IPYNB script to read the data of the following matrix (stored in CSV, JSON, or any other format/data structure that you choose) and implement the calculations needed for the following requirements.

Consider the data listed in the following matrix for a product of size 120 KLOC and provide your answers for the following requirements:

1. Calculate the defect removal effectiveness for every inspection phase
2. Calculate the defect removal effectiveness for every testing phase
3. Calculate the overall defect removal effectiveness for the process.
4. Create a stacked bar chart to show the Defects Originated, Detected, and Escaped for every phase.

Resources/Tutorials:

- https://seaborn.pydata.org/tutorial/objects_interface.html
- <https://python-graph-gallery.com/stacked-and-percent-stacked-barplot/>

Defect Origin									
Where Found		Requirement	Analysis	Design	Coding	Unit Testing	Integration Testing	System Testing	Field
	Requirement	52							
	Analysis	18	31						
	Design	17	17	78					
	Coding	36	19	55	189				
	Unit Testing	12	43	8	156	27			
	Integration Testing	17	23	26	45	-	3		
	System Testing	24	39	57	41	-	-	2	
	Field	2	4	3	5	-	-	-	5

PART II (100 points):

In this part you will use GenAI model gpt-4o-mini to automate the evaluation of the defect removal effectiveness by using LangGraph and gpt-4o-mini to generate and execute python code.

Implement the requirements in Part I using Python, Langchain/Langgraph and OpenAI/gpt-4-mini. The AI Agent representing the Software Project Manager shall use GenAI model to generate Python code and us Python REPL to execute the generated Python code.

Resources/Tutorials:

- <https://langchain-opentutorial.gitbook.io/langchain-opentutorial/17-langgraph/03-use-cases/07-langgraph-multi-agent-supervisor>
- https://python.langchain.com/api_reference/experimental/agents/langchain_experimental.agents.agent_toolkits.csv.base.create_csv_agent.html
- https://python.langchain.com/docs/how_to/sql_csv/
- <https://medium.com/mitb-for-all/a-gentle-introduction-to-the-llm-multi-agents-multiverse-part-1-langgraph-2ac56f1b5b3c>
- <https://python.langchain.com/docs/integrations/tools/python/>
- <https://github.com/langchain-ai/data-enrichment>